ME101: Engineering Mechanics, 2023-24

IIT Guwahati

Tutorial – 10

Tutorial Date: 8th April 2024 (7:55 am – 8:50 am)

Problem 1:

The bent rod *ABCDE* rotates about a line joining Points *A* and *E* with a constant angular velocity of 20 rad/s. Knowing that the rotation is clockwise as viewed from E, determine the velocity and acceleration of corner *C*. Assume the dimensions are a = 40mm, b = 50mm, c = 30mm, d = 100mm, e = 30mm. [10 Marks]

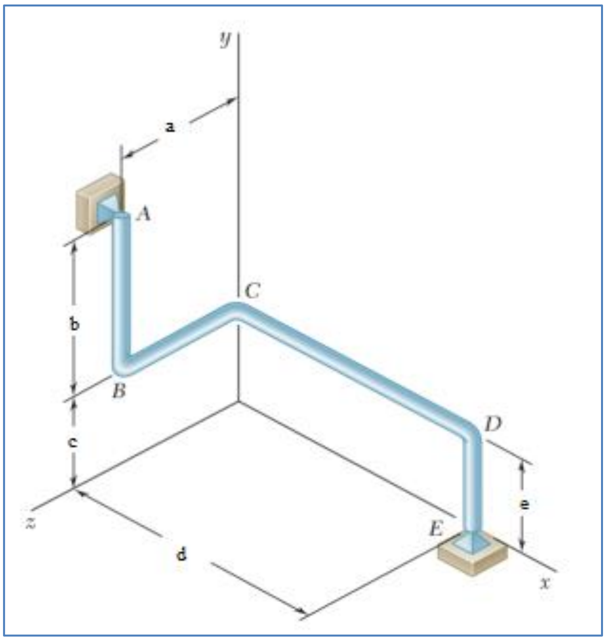

Figure 1: Problem 1

Problem 2:

Ring *C* has an inside radius of 55 mm and an outside radius of 60 mm, here t is 5 mm and is positioned between two wheels *A* and *B*, each of 24-mm (R) outside radius. Knowing that wheel A rotates with the clockwise angular velocity of 300 rpm which reduces at the rate of 20 rad/s^2 and that no slipping occurs, determine (a) the angular velocity of the ring *C* and wheel *B*, (b) the acceleration of the points on *A* and *B* that are in contact with *C*. [10 Marks]

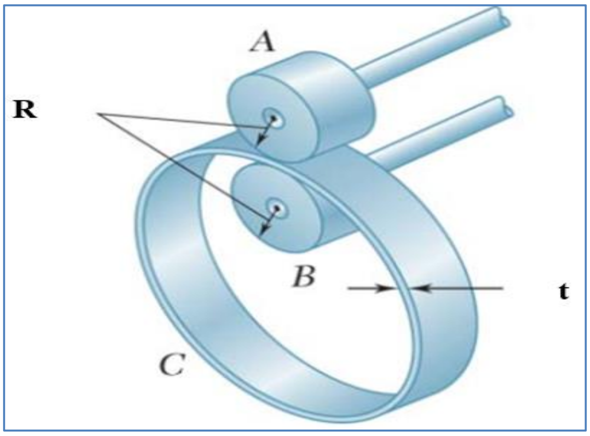

Figure 2: Problem 2

Problem 3:

Activation of the hydraulic cylinder causes *OB* to elongate at the constant rate of 0.5 m/s. Determine both tangential and normal acceleration of point A in its circular path around

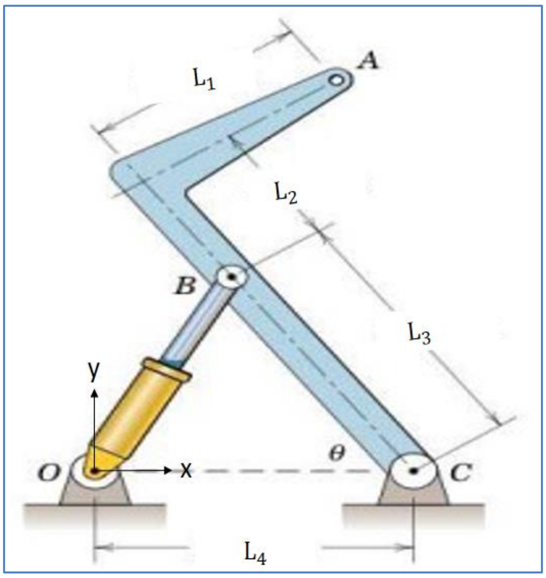

Figure 3: Problem 3

C for the instant when $\theta = 60^o$ and for $L_1 = 375$ mm, $L_2 = 250$ mm, $L_3 = 500$ mm, $L_4 = 500$ mm. [10 Marks]

Problem 4:

The cable from drum A turns the double wheel B, which rolls on its hubs without slipping. Determine the angular velocity ω and angular acceleration α of drum C for the instant when the angular velocity and angular acceleration of A are $5\ \text{rad/s}$ and $4\ \text{rad/s}^2$, respectively, both in the counter clockwise direction. Take $r_A = r_C = 0.1$ m, $r_o = 0.4$ m and $r_i = 0.2$ m.

[10 Marks]

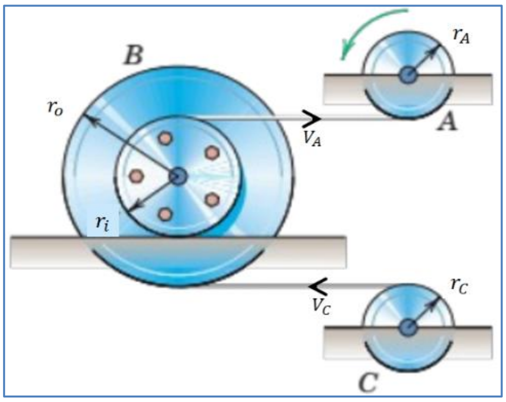

Figure 4: Problem 4

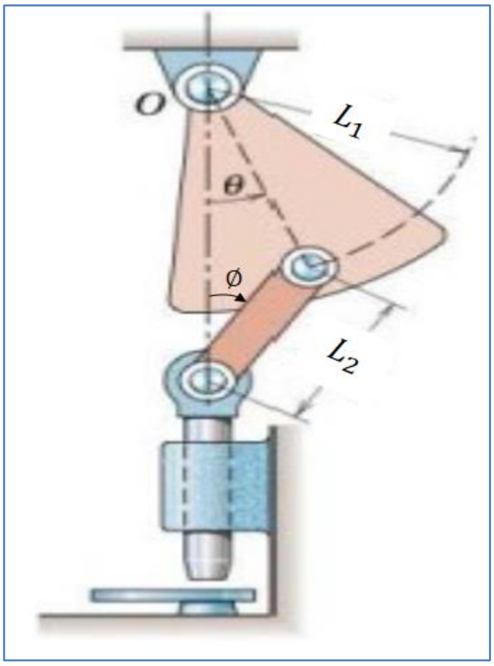

Figure 5: Problem 5

Problem 5:

The punch (Fig. 5) is operated by a simple harmonic oscillation of the pivoted sector given by $\theta = \theta_0 \sin 2\pi t$ where the amplitude is $\theta_0 = \pi/10$ rad (18°) and the time for one complete oscillation is 1 second. Determine the acceleration of the punch when $(a)\ \theta = 0$ rad and $(b)\ \theta = \frac{\pi}{12}$ rad, considering $L_1 = 300$ mm and $L_2 = 200$ mm. [10 Marks]